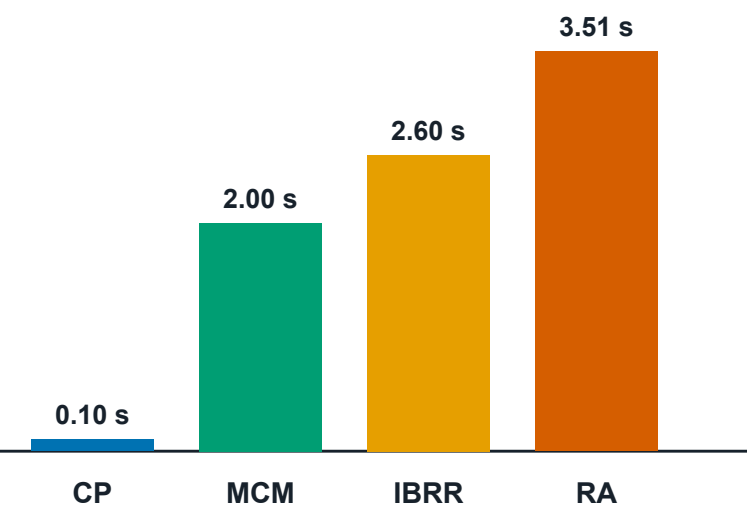


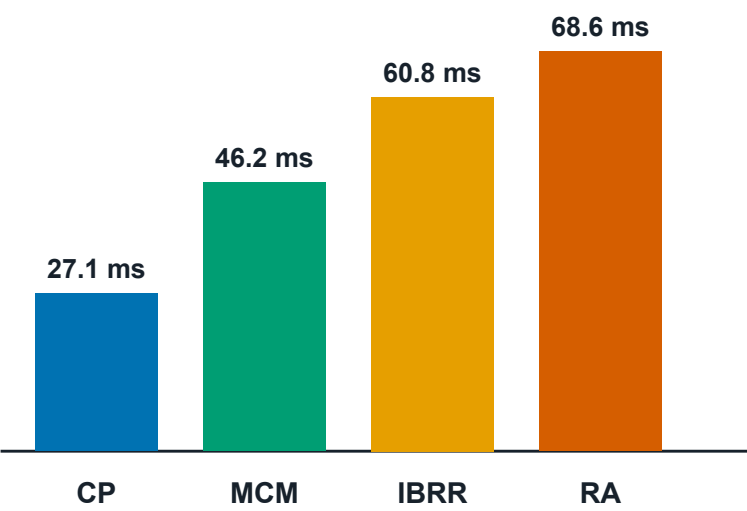
Six views of the staged variants

Median timing and fixed proof metrics at d = 13; n = 30 independent warmed processes

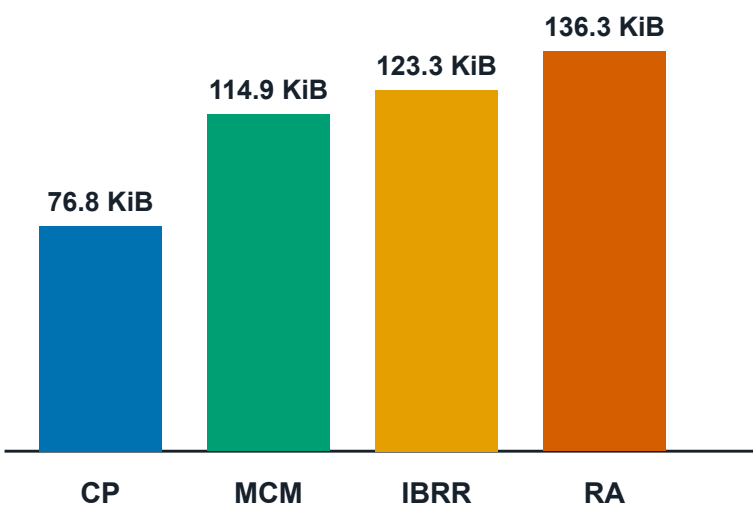
A. Prover time



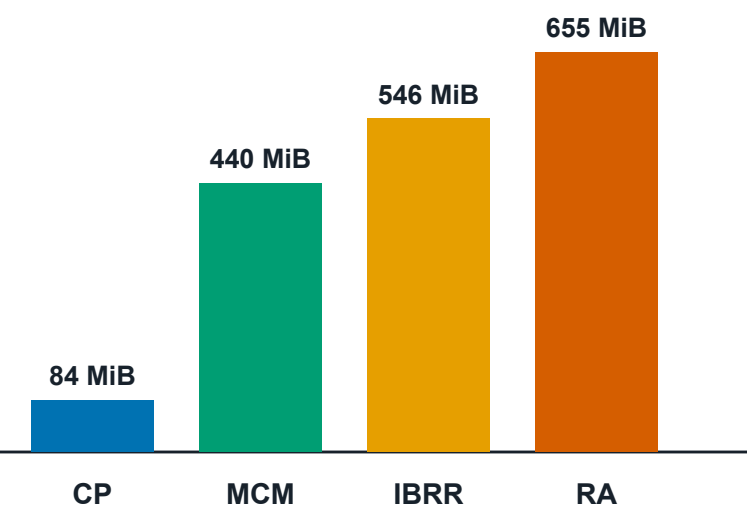
B. Verifier time



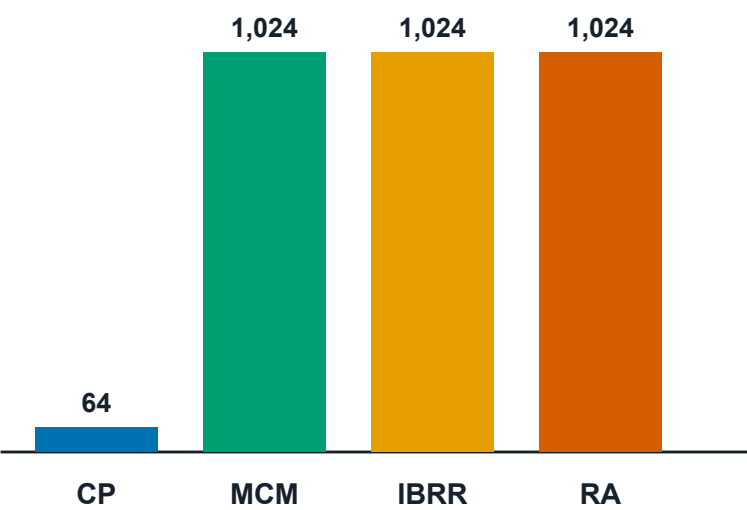
C. Proof size



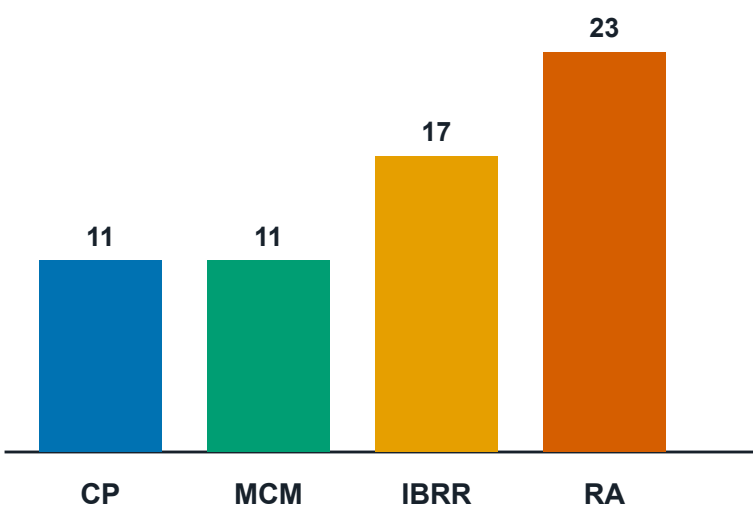
D. Observed peak RSS



E. Execution trace



F. AIR transition constraints



Memory is maximum observed resident set after proving; transition constraints are AIR column constraints, not R1CS rows.